

- 3 Carrots are root vegetables of the carrot plant. The carrot plant is an important food crop that is grown throughout the world. Carrots have a sweet taste because sugars form a proportion of the total carbohydrate present.

Plant breeding has produced many different varieties of carrot, with different levels of sweetness.

- (a) One of the sugars found in carrots is galactose.

Galactose has the same molecular formula, $C_6H_{12}O_6$, as α -glucose.

Fig. 3.1 shows the molecular structure of galactose found in carrots. This is similar, but not identical, to the molecular structure of α -glucose.

In Fig. 3.1, the six carbon atoms are numbered 1 to 6.

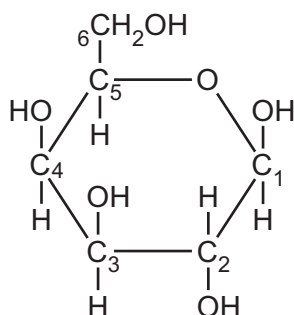


Fig. 3.1

State the differences between the molecular structure of galactose, shown in Fig. 3.1, and the molecular structure of α -glucose.

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..... [1]

- (b) An investigation was carried out to determine the sugar content of mature carrots produced by different local varieties of carrot plants that are grown in Tunisia, North Africa.

Carrot plants were grown from seed under standardised conditions. When mature, the carrots were harvested.

The different sugars present in the carrots were identified. Measurements of sugar content for each of the sugars were made.

Table 3.1 shows the results for five different local varieties of carrot plant, **A** to **E**.

Table 3.1

local variety	sugar content of carrots /mg g ⁻¹ of dry weight			
	fructose	galactose	glucose	sucrose
A	183.04	1.43	173.28	73.73
B	194.43	1.13	188.18	46.20
C	200.15	5.99	125.19	85.93
D	157.35	4.88	137.28	89.57
E	170.38	4.54	133.57	60.27

After studying the results shown in Table 3.1, a student concluded that the carrots from the local varieties contain the same four sugars.

The student made three other conclusions from the data in Table 3.1.

- conclusion 1 There are non-reducing and reducing sugars in the carrots.
- conclusion 2 There are monosaccharide and disaccharide sugars in the carrots.
- conclusion 3 The carrots have the same pattern of results.



Explain the evidence in Table 3.1 that supports these three other conclusions.

conclusion 1

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conclusion 2

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conclusion 3

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[3]

(c) Carbohydrates that are **not** sugars are also present in carrots.

Name **one** carbohydrate that is present in carrots and that is **not** a sugar.

..... [1]

- (d) Each carrot plant produces a carrot in the first year of growth.

If the carrot is **not** removed from the plant after it matures, the plant passes through a dormant period and shoots develop from the carrot in the second year of growth. This allows flowers to be produced and seed formation to occur before the plant dies.

Explain, with reference to the life cycle of the carrot plant, when the carrot acts as a **source** and when the carrot acts as a **sink**.

source

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sink

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[3]

- (e) Carrot virus Y is a pathogen of carrot plants. The virus, which belongs to a group known as *Potyvirus*, replicates its viral nucleic acid and proteins within host carrot cells.

The general structure of potyviruses is shown in Fig. 3.2.

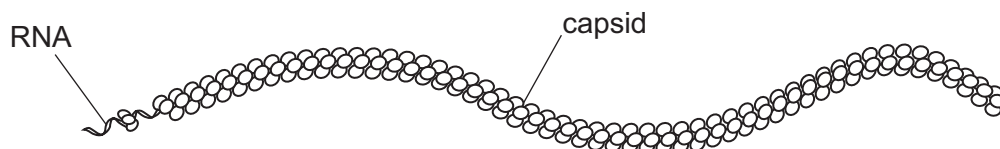


Fig. 3.2

The synthesis of viral proteins in host carrot cells only involves the process of translation. The process of transcription does **not** occur.

Suggest why translation occurs in host carrot cells during the synthesis of **viral** proteins, but transcription does **not** occur.

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[2]

[Total: 10]